

Lesson 4

The compensated law of demand, the Slutsky matrix, and the limits of WARP

1 The compensated law of demand, and WARP

The compensated law of demand is an “if and only if” statement: for demand that is homogeneous of degree 0 and satisfies Walras’ law, its two parts are equivalent, so WARP says the same thing as

$$(P' - P) \cdot (x(P', w') - x(P, w)) \leq 0 \quad \text{whenever} \quad w' = P' \cdot x(P, w),$$

with strict inequality whenever $x(P', w') \neq x(P, w)$. That is, when the price change is accompanied by the wealth compensation that keeps the original bundle just affordable, the compensated change in demand points in the direction opposite to the price change. What follows proves the implication $\text{WARP} \Rightarrow$ the inequality.

2 Proof: WARP implies the compensated law of demand

The trivial case

If $x(P', w') = x(P, w)$ then $x(P', w') - x(P, w) = 0$, so the left-hand side is zero and the inequality holds.

The case $x(P', w') \neq x(P, w)$

By construction of the compensated wealth, $P' \cdot x(P, w) = w'$, and by Walras’ law at the new prices the agent spends all of it, $w' = P' \cdot x(P', w')$. Combining the two,

$$P' \cdot (x(P', w') - x(P, w)) = 0. \quad (*)$$

Now use WARP. Since $P' \cdot x(P, w) = w'$, the old bundle $x(P, w)$ was affordable when $x(P', w')$ was chosen, so

$$x(P', w') \text{ is revealed preferred to } x(P, w).$$

The opposite revealed preference cannot also hold, for that would be a self-contradiction. If $x(P', w')$ had been affordable at (P, w) , then — $x(P, w)$ being the bundle chosen there — $x(P, w)$ would be revealed preferred to $x(P', w')$, and the two bundles would each be revealed preferred to the other. WARP rules this out, so $x(P', w')$ was *not* affordable at (P, w) :

$$P \cdot x(P', w') > w.$$

Subtracting $w = P \cdot x(P, w)$ — Walras’ law at the original prices — from this,

$$P \cdot (x(P', w') - x(P, w)) > 0.$$

Finally, use (*):

$$\begin{aligned} (P' - P) \cdot (x(P', w') - x(P, w)) &= P' \cdot (x(P', w') - x(P, w)) - P \cdot (x(P', w') - x(P, w)) \\ &= 0 - P \cdot (x(P', w') - x(P, w)) < 0, \end{aligned}$$

which is the compensated law of demand, strict in this case.

3 The marginal price change: the substitution effect

Now consider a small change in the price p_j alone, with the wealth compensation applied at the same time. The question is: by how much does the demand for good i respond to p_j when wealth is adjusted so that the *old* bundle stays just affordable? That response is the substitution effect S_{ij} .

Step 1: fix the starting point, and write compensated demand as a function of p'

Assume demand is differentiable. Start at prices p and wealth w , where the consumer buys $x(p, w)$. Treat (p, w) , and therefore the old bundle $x(p, w)$, as *fixed numbers*: they describe where we start and do not move. The only variable is the new price vector p' . Slutsky compensation sets wealth at the new prices to

$$w'(p') = p' \cdot x(p, w) = \sum_{k=1}^l p'_k x_k(p, w),$$

so the compensated demand for good i is a function of p' alone,

$$h_i(p') \equiv x_i(p', w'(p')) = x_i(p', p' \cdot x(p, w)).$$

Step 2: see where p'_j enters

In $x_i(p', w'(p'))$ the new price p'_j enters in two places:

1. *directly*, as the j -th price argument of the demand function;
2. *through wealth*, since $w'(p') = \sum_k p'_k x_k(p, w)$. Because the old bundle $x(p, w)$ is a constant here, only the $k = j$ term depends on p'_j , so

$$\frac{\partial w'}{\partial p'_j} = x_j(p, w) :$$

raising p'_j by one unit raises the compensation by exactly the amount of good j in the old bundle.

Step 3: the chain rule

Adding the two channels,

$$\frac{\partial h_i(p')}{\partial p'_j} = \underbrace{\frac{\partial x_i}{\partial p'_j}(p', w')}_{\text{direct price effect}} + \underbrace{\frac{\partial x_i}{\partial w}(p', w') \frac{\partial w'}{\partial p'_j}}_{\text{wealth effect}} , \quad w' = p' \cdot x(p, w).$$

$= x_j(p, w)$

Here $\partial x_i / \partial p'_j$ is the partial derivative of the demand function $x_i(\cdot, \cdot)$ with respect to its j -th price argument, holding its wealth argument fixed, and $\partial x_i / \partial w$ its derivative with respect to the wealth argument; both are evaluated at the new point (p', w') . Note that $x_j(p, w)$ carries the *old* prices: it comes from $\partial w' / \partial p'_j$, and the old bundle does not move.

Step 4: evaluate at $p' = p$

S_{ij} is the effect of an *infinitesimal* change, so the derivative is taken at the starting point, $p' = p$. There the compensated wealth is $w' = p \cdot x(p, w) = w$ by Walras' law, so the evaluation point (p', w') is just (p, w) , and the price argument written p'_j is then p_j . This gives the *substitution effect* of a change in the price of good j on good i :

$$S_{ij} \equiv \left. \frac{\partial h_i(p')}{\partial p'_j} \right|_{p'=p} = \frac{\partial x_i(p, w)}{\partial p_j} + \frac{\partial x_i(p, w)}{\partial w} x_j(p, w).$$

Reading the formula

The ordinary (uncompensated) price effect $\partial x_i / \partial p_j$ mixes two things: relative prices change, *and* the consumer becomes poorer (by x_j per unit rise in p_j , to first order). The compensation hands back exactly that loss, and the term $\frac{\partial x_i}{\partial w} x_j$ is the demand response to it. What is left, S_{ij} , is the response to the change in relative prices alone. Rearranged, this is the *Slutsky equation*:

$$\underbrace{\frac{\partial x_i(p, w)}{\partial p_j}}_{\text{total effect}} = \underbrace{S_{ij}}_{\text{substitution effect}} - \underbrace{\frac{\partial x_i(p, w)}{\partial w} x_j(p, w)}_{\text{wealth effect}}.$$

4 The Slutsky matrix

The $l \times l$ matrix that collects all the S_{ij} is the *substitution matrix*, or *Slutsky matrix*. What can be said about its structure? The clue is the compensated law of demand, which in differential form reads

$$\sum_{i=1}^l dp_i dx_i \leq 0.$$

The change in wealth that accompanies the price change is

$$dw = x(P, w) \cdot dP = \sum_{j=1}^l x_j(P, w) dP_j,$$

so the total differential of the demand for good i is

$$\begin{aligned} dx_i &= \sum_{j=1}^l \frac{\partial x_i(P, w)}{\partial P_j} dP_j + \frac{\partial x_i(P, w)}{\partial w} dw \\ &= \sum_{j=1}^l \frac{\partial x_i(P, w)}{\partial P_j} dP_j + \frac{\partial x_i(P, w)}{\partial w} \sum_{j=1}^l x_j(P, w) dP_j \\ &= \sum_{j=1}^l \left[\frac{\partial x_i(P, w)}{\partial P_j} + \frac{\partial x_i(P, w)}{\partial w} x_j(P, w) \right] dP_j \\ &= \sum_{j=1}^l S_{ij} dP_j. \end{aligned}$$

5 Negative semi-definiteness of the Slutsky matrix

Substituting $dx_i = \sum_j S_{ij} dP_j$ into the compensated law of demand,

$$\sum_{i=1}^l dP_i dx_i = \sum_{i=1}^l dP_i \sum_{j=1}^l S_{ij} dP_j = \sum_{i=1}^l \sum_{j=1}^l dP_i S_{ij} dP_j \leq 0,$$

by the compensated law of demand. Since this holds for every price change dP , the quadratic form $dP' S dP$ is non-positive for every dP : the Slutsky matrix is *negative semi-definite*, in the sense $v' S v \leq 0$ for all $v \in \mathbb{R}^l$, which does not require S to be symmetric.

Later in the course the same conclusion is obtained when axioms are imposed on preferences. (The quadratic form only pins down the symmetric part $\frac{1}{2}(S + S')$. WARP does not deliver symmetry of S in general — with two goods S is automatically symmetric, given homogeneity and Walras' law, but not with three or more — whereas the preference-based derivation also gives symmetry.)

S is never negative *definite*. By Walras' law and then homogeneity of degree 0 (the Euler equation of Lesson 3),

$$\sum_j S_{ij} p_j = \sum_j \frac{\partial x_i}{\partial p_j} p_j + \frac{\partial x_i}{\partial w} p \cdot x(p, w) = \sum_j \frac{\partial x_i}{\partial p_j} p_j + \frac{\partial x_i}{\partial w} w = 0,$$

so $S p = 0$ and $p' S p = 0$: a proportional change in all prices, fully compensated, changes nothing.

6 WARP alone is not enough: an example

Consider three price situations and the bundles chosen in each:

$$\begin{aligned} P^0 &= (1, 1, 2), & P^1 &= (1, 1, 1), & P^2 &= (1, 2, 1), \\ x^0 &= (5, 19, 9), & x^1 &= (12, 12, 12), & x^2 &= (27, 11, 1). \end{aligned}$$

Here $x^i = x(P^i, w^i)$, so the wealth in situation i is what x^i costs at P^i . The cost of each bundle at each price vector is

	P^0	P^1	P^2
x^0	42	33	52
x^1	48	36	48
x^2	40	39	50

the bold entry in each column being the wealth, i.e. the cost of the bundle actually chosen there: $w^0 = 42$, $w^1 = 36$, $w^2 = 50$. Reading the columns:

- At P^0 (wealth 42): x^1 costs 48 and is not affordable, but x^2 costs 40. So x^2 was in the budget and x^0 was chosen: x^0 R.P. x^2 .
- At P^1 (wealth 36): x^2 costs 39 and is not affordable, but x^0 costs 33, so x^1 R.P. x^0 .
- At P^2 (wealth 50): x^0 costs 52 and is not affordable, but x^1 costs 48, so x^2 R.P. x^1 .

Chaining the three,

$$x^0 \text{ R.P. } x^2 \text{ R.P. } x^1 \text{ R.P. } x^0,$$

a cycle: the revealed preferences are *intransitive*. And yet WARP is satisfied at every pair. In each of the three comparisons above the alternative bundle is affordable only *one* way round — x^0 is not affordable at P^2 , x^1 is not affordable at P^0 , and x^2 is not affordable at P^1 — so no two bundles are ever revealed preferred to each other directly. WARP, which only ever compares two situations at a time, never notices the cycle.

So WARP on its own is not enough to establish that the agent has rational (that is, transitive) preferences. The example needs three goods: with only two goods, WARP does rule out such cycles.

7 The strong axiom of revealed preference (SARP)

What is missing is transitivity. The *strong axiom of revealed preference* adds it: no chain of revealed preferences may return to its starting point,

$$x^0 \text{ R.P. } x^1 \text{ R.P. } \dots \text{ R.P. } x^0 \text{ is forbidden.}$$

SARP extends WARP from pairs to chains of any length (a chain of length two is exactly what WARP forbids), so SARP implies WARP but not conversely; the example above satisfies WARP and fails SARP, which is exactly why it is the counterexample it is.

Transcription notes

- The original opens with a title page reading only “Lesson 4”.
- **Corrected:** in the total differential of dx_i the original sums over i inside the sum over i and then keeps the index i on the marginal effect of wealth, “ $\sum_j \frac{\partial x_i}{\partial P_j} dP_j + \frac{\partial x_i}{\partial w} \sum_i x_i(P, w) dP_i$ ”. The second sum must run over the good whose price changed, and the term multiplying $\frac{\partial x_i}{\partial w}$ is $x_j(P, w) dP_j$ — which is what makes the bracket come out as S_{ij} on the next line. Both places are written with j here.
- **Corrected:** the derivative that defines the substitution effect is taken with respect to p'_j (the price inside the compensated demand), not with respect to p_j ; the notation p' is used throughout.
- **Corrected:** the last page reads “Strong axiom A.P. (SARP)”; the axiom is the strong axiom of *revealed preference*. The line reads “Assume away transitivity cycle” (“away” inserted above the line), i.e. SARP assumes away cycles of revealed preference — the intransitive cycles of the example.
- Negative semi-definite: the notes conclude straight from $\sum_i \sum_j dP_i S_{ij} dP_j \leq 0$ that the Slutsky matrix is negative semi-definite, and that conclusion is correct as it stands, with semi-definiteness meant as $v' S v \leq 0$ for all v (as in MWG), which does not presuppose symmetry. **Corrected:** an earlier version of this transcription said the conclusion needed symmetry; the remark now says only that the quadratic form sees the symmetric part, and that symmetry is a separate property, obtained later from axioms on preferences.
- Page 5’s remark “Later, we get the same conclusion when we impose axioms on preferences” is kept as the teacher’s statement (it refers to negative semi-definiteness). **Corrected:** an earlier version of this transcription turned it into a claim that the *next lecture* derives symmetry. **Added:** the parenthetical on symmetry.
- **Added:** the result $Sp = 0$ (so S is never negative definite), the assumptions (homogeneity, Walras’ law) under which the “if and only if” holds, the strict inequality when the bundles differ, and the remark that the WARP-without-SARP example needs three goods.
- **Corrected:** in the example, “both were in the budget” at P^0 said x^1 was affordable there; it costs 48 against a wealth of 42, and only x^2 was in the budget.

- The chain-rule line wrote every partial derivative with the same composite argument $x_i(p', p' \cdot x(p, w))$, which makes the right-hand side look like the left-hand side. **Corrected:** an earlier version of this transcription wrote the direct term as $\partial x_i / \partial p_j$ while differentiating in p'_j ; since the derivative is with respect to the new price, the direct term is $\partial x_i / \partial p'_j$, evaluated at (p', w') . It becomes $\partial x_i(p, w) / \partial p_j$ only after evaluating at $p' = p$. **Added:** the derivation is split into four steps (fix the starting point; the two channels through which p'_j enters; the chain rule; evaluation at $p' = p$ using Walras' law), with the compensated demand named $h_i(p')$, and the Slutsky equation is stated.
- “slatsky” is spelled *Slutsky* here, and “intransive” is *intransitive*.
- The original uses lower case p in the substitution-effect section and upper case P in the Slutsky-matrix section; the case is kept as written, as elsewhere in these notes.
- The example is transcribed with its numbers unchanged; the cost table was recomputed from the price vectors and bundles and matches the original entry by entry (x^0 : 42, 33, 52; x^1 : 48, 36, 48; x^2 : 40, 39, 50). The original writes the table with the bundles as rows and the price vectors as columns, which is kept.
- The line that introduces the marginal-price-change section reads “we move to key results readings marginal price change” in the original; the word after “results” is probably “regarding”. The sentence itself is not typeset; the section heading replaces it.
- **Corrected:** the original introduces the marginal change as “change in x_i when P_i changes”; the derivation that follows (and S_{ij}) concerns a change in the price p_j of good j , so p_j is used.
- On the proof page, the original’s step “ $x(P', w')$ is R.P. to $x(P, w)$, so WARP tells us that $x(P, w)$ was picked under (P, w) because $x(P', w')$ was not affordable” compresses the argument: that $x(P', w')$ was not affordable at (P, w) is the *conclusion* drawn from WARP (the opposite revealed preference would be a contradiction), not a premise. The proof is written out with that step spelled out.
- Pages of the original, for tracing back: page 2 the “iff” statement and the proof of $\text{WARP} \Rightarrow$ the inequality; page 3 the marginal price change and S_{ij} ; page 4 the Slutsky matrix and the differential form; page 5 the negative semi-definiteness conclusion; page 6 the three-price example; page 7 the strong axiom.